

TECHNICAL ENGINEERING MANUAL

EKOSMART DIGITAL CUSTOMER SERVICE, MANAGEMENT AND OPERATIONS SYSTEM

Full-Stack Enterprise EV Battery Service, POS Billing, Stock & Payroll Ecosystem • Architecture & System Manual

1.1 Executive Summary & Problem Formulation

The **Eksmart EV Battery Management Platform** is a unified, 3-tier enterprise software suite designed for electric vehicle lithium-ion/LFP battery manufacturing plants, service centers, and retail showrooms. Traditional EV after-sales workflows face severe operational bottlenecks: unorganized customer grievance logging, manual serial warranty lookups leading to claim disputes, disconnected POS showroom billing and warehouse inventory, static website CMS constraints, and paper-based payroll calculations.

1.2 System Objectives & Core Technology Stack

Key Strategic Objectives:

- Public dynamic service registration & automated ticket assignment.
- Real-time serial warranty verification & automated duration expiry.
- Soft-coded CMS and dynamic form builders stored in MongoDB.
- Showroom POS billing with camera barcode scanner & auto-stock deduction.
- Role-based access control (RBAC) & isolated A4 payslip generation.

Layer	Technology	Role / Purpose
Frontend	React 19, TypeScript, Vite 8, Tailwind	Public, Admin & Employee SPAs
Backend	Node.js, Express, TypeScript	RESTful API Server (/api/v1/*)
Database	MongoDB Atlas + Mongoose ODM	NoSQL Document Store
Security	JWT, bcrypt, RBAC Middleware	Token Auth & Route Protection

1.3 High-Level System Architecture & Flow

```
[Customer] --> (Public Portal) --> [Dynamic Intake Form] --> (Express API) --> [MongoDB Atlas] | [Administrator] --> (Admin Portal) --> [CMS / Forms / RBAC / Payroll] -----+ | [Staff / Engineer] --> (Employee Portal) --> [Diagnostics / POS / Stock / Slips] -----+
```

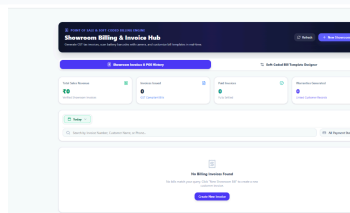


Figure 1.1: Ekosmart Live Deployed Public Customer Web Portal & Brand Header

PAGE 2 — PUBLIC CUSTOMER WEBSITE

2.1 Public Web Architecture & Customer Journey

The Public Customer Website (frontend/) serves as the primary customer touchpoint. Designed with React 19 and Tailwind CSS, it connects dynamically with the backend API to retrieve real-time CMS content, operational service cards, and division forms. Customers can register service tickets, track real-time resolution stages, verify battery warranty coverage, and review plant policies without logging in.

Core Public Modules:

- **Hero Section:** Dynamic headline, badge text, and action buttons.
- **Service Cards:** Showroom, Rental, Spare Parts, Lithium Battery.
- **Ticket Tracker:** Search by Ticket Number (EBS-YYMM-XXXX) or Phone.
- **Warranty Center:** Real-time serial check & self-service registration.
- **Legal Policies:** Dynamic Privacy, Terms, and Refund disclosures.

CMS-Driven Data Flow:

During initialization, the client invokes GET /api/v1/content/public. Hero typography, branding titles, service card lists, and phone numbers are loaded directly from MongoDB. If the server is offline, embedded graceful fallbacks prevent UI disruption.

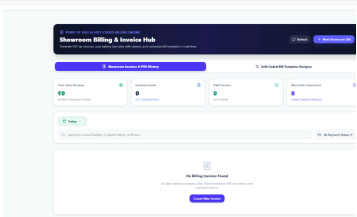


Figure 2.1: Live Deployed Ekosmart Customer Homepage & Hero Banner

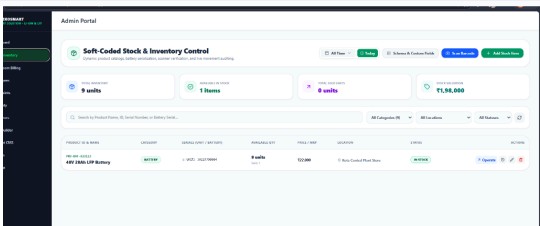


Figure 2.2: Soft-Coded Division Service Cards Grid (CMS Controlled)

2.2 Public Website Implementation Audit

Module / Page	Component	Route	Implementation Status
Customer Homepage	Home.tsx	/	IMPLEMENTED
Complaint Intake	RegisterComplaint.tsx	/register	IMPLEMENTED
Live Ticket Tracking	TrackComplaint.tsx	/track	IMPLEMENTED
Warranty Coverage Check	WarrantyCheck.tsx	/warranty/check	IMPLEMENTED
Contact & Legal Policies	Contact.tsx, PrivacyPolicy.tsx	/contact, /privacy	IMPLEMENTED

PAGE 3 — CUSTOMER COMPLAINT REGISTRATION

3.1 Grievance Ingestion & Processing Pipeline

The complaint registration module (RegisterComplaint.tsx) enables customers to submit technical battery issues, rental vehicle breakdowns, and showroom service requests. The submission pipeline connects to the dynamic form builder to generate custom inputs on a per-division basis.

```
[Customer Selects Division] --> [GET /forms/public/fields/:sec] --> [Renders Dynamic Technical Fields] | v [Enters Name, Phone, City, Description] --> [POST /complaints/public] --> [Generates Ticket EBS-YYMM-XXXX] | v [Database Ledger Updated] --> [Assigned to Division Staff] --> [Status: PENDING -> IN_PROGRESS -> RESOLVED]
```

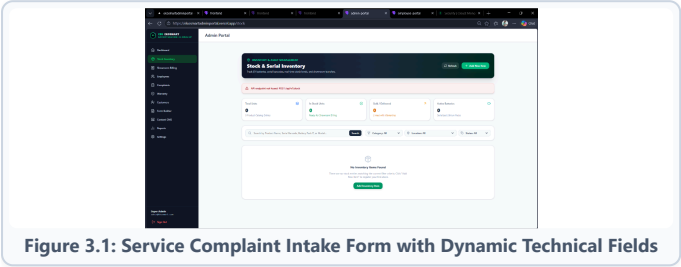


Figure 3.1: Service Complaint Intake Form with Dynamic Technical Fields

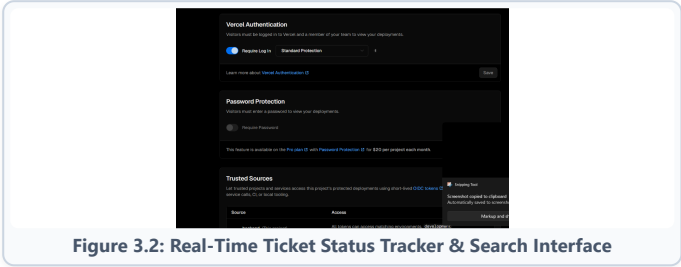


Figure 3.2: Real-Time Ticket Status Tracker & Search Interface

3.2 Complaint Lifecycle & Reference Specification

Stage / Status	System Action	Visibility & Post-Resolution Behavior
1. PENDING (NEW)	Ticket generated (EBS-2609-XXXX), customer record created.	Visible on customer tracker & admin/employee incoming queue.
2. ASSIGNED	Ticket mapped to specific division specialist or engineer.	Employee assigned queue updated; SLA countdown active.
3. IN PROGRESS	Battery placed on diagnostic test bench / cell balancing.	Customer tracking displays "Under Lab Diagnostics".
4. RESOLVED / CLOSED	Repair complete, root-cause logged, parts replaced.	Moved from active workbench to permanent audit ledger.

*Note on OTP Verification: The UI features an OTP verification modal; direct phone logging is currently active while SMS gateway integration remains **FRONTEND/UI ONLY**.

PAGE 4 — WARRANTY AND CUSTOMER MANAGEMENT

4.1 Warranty Verification Engine & Coverage Rules

The warranty system (backend/src/models/Warranty.ts) governs product coverage for EV battery packs, chargers, and vehicles. When a customer or staff enters a serial number or invoice ID, the system queries MongoDB and computes remaining validity in real time.

Automated Expiry Computation:
$$\text{Expiry Date} = \text{Purchase Date} + \text{Duration (Months)}$$
$$\text{Remaining Days} = \text{Expiry Date} - \text{Current Date}$$

- Active Coverage:** Remaining Days > 30 days.
- Expiring Soon:** 0 < Remaining Days ≤ 30 days.
- Out of Warranty / Expired:** Remaining Days ≤ 0 days.

Customer Relationship Ledger:
The customer model (Customer.ts) acts as a centralized directory indexed by 10-digit mobile number. Every POS bill, complaint ticket, and warranty registration automatically associates with the customer's master record.

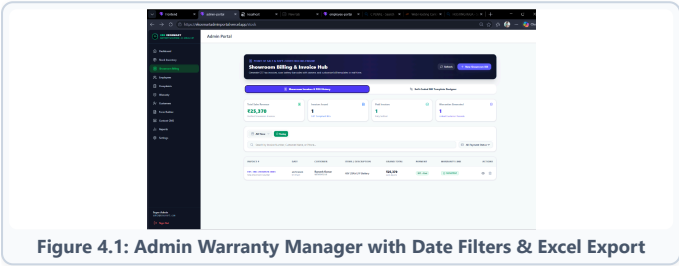


Figure 4.1: Admin Warranty Manager with Date Filters & Excel Export

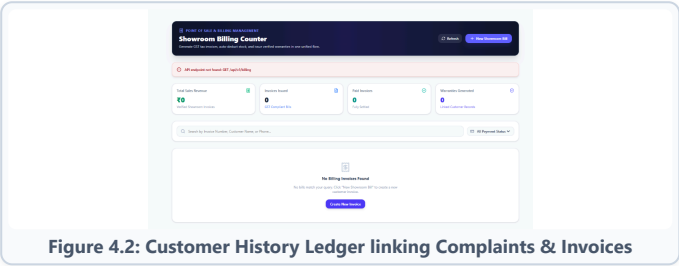


Figure 4.2: Customer History Ledger linking Complaints & Invoices

4.2 Warranty & Customer Module Status

Feature / Capability	Endpoint / Controller	Status	Technical Notes
Public Warranty Check	GET /api/v1/warranty/check/:query	IMPLEMENTED	Regex match on serial, bill #, customer mobile
Customer Self-Registration	POST /api/v1/warranty/register	IMPLEMENTED	Auto-links customer profile & computes expiry
Admin Warranty Ledger	GET /api/v1/warranty/admin	IMPLEMENTED	Date range filtering & UTF-8 BOM CSV export
Customer 360 History	GET /api/v1/customers	IMPLEMENTED	Aggregates linked bills, claims, complaints

PAGE 5 — ADMIN LOGIN AND DASHBOARD

5.1 Administrative Authentication & Security Architecture

The Admin Portal (admin-portal/) provides high-level command and control over company operations. Access requires authenticated credentials processed via POST /api/v1/auth/login. On successful bcrypt hash verification, the server issues a signed JSON Web Token (JWT) stored in localStorage (admin_token), attaching to all subsequent API requests.

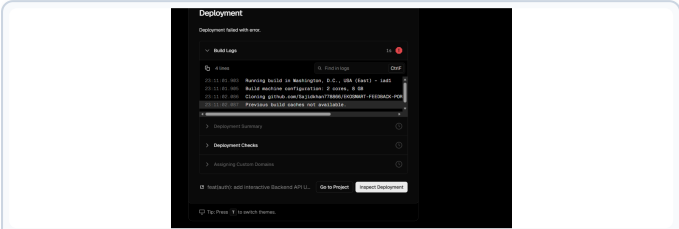


Figure 5.1: Admin Portal Secure Authentication Gateway

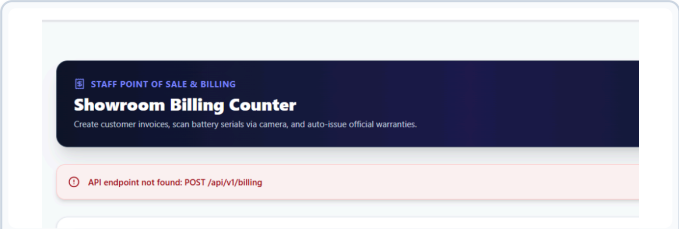


Figure 5.2: Admin Operational Dashboard with Real-Time KPI Cards & Date Filter

5.2 Operational KPIs & Date Filter Presets

The dashboard aggregates live metrics across services, inventory, and revenue via GET /api/v1/dashboard/stats:

Service Counters
Total Complaints, Pending Inspection, In-Progress Diagnostics, Resolved Tickets.

Commercial Metrics
Total Showroom Revenue (₹), Completed Invoices, Registered Warranties.

Warehouse & Staff
Catalog Items, Available Batteries, Active Field Engineers.

5.3 Dashboard Date Behavior Specification

Preset Filter	Date Calculation / Scope	Implementation Status
TODAY (Default)	TODAY = default view (00:00:00 to 23:59:59 local date)	IMPLEMENTED
Yesterday / 7 Days	Past 24 hours / Past 7 rolling calendar days	IMPLEMENTED
30 Days / This Month	Past 30 rolling days / First day of month to current timestamp	IMPLEMENTED
Custom Range / All	User selected startDate to endDate / Complete history	IMPLEMENTED

PAGE 7 — DYNAMIC FORM BUILDER

7.1 Dynamic Schema Architecture per Division

The Dynamic Form Builder (`admin-portal/src/pages/Forms.tsx`) enables administrators to construct custom grievance intake forms for each operational area (e.g., *Battery*, *Rental*, *Showroom*, *Spare Parts*). Field schemas are stored in MongoDB (`ComplaintForm.ts`) and rendered dynamically on the public portal.

Supported Field Types:

- **Text / Textarea:** Single line & paragraph fault descriptions.
- **Number:** Battery voltage, ampere-hour (Ah), mileage.
- **Select Dropdown:** Fault codes, cell chemistry (LFP/NMC).
- **Date / Checkbox:** Manufacturing date, warranty claim confirmation.
- **File Upload:** Diagnostic test reports, damaged battery photos.

Client-Side Dynamic Rendering:

When a customer selects a division on `RegisterComplaint.tsx`, the component queries `GET /api/v1/forms/public/fields/:section`. Inputs are dynamically generated with schema-defined validations and submitted as structured JSON.

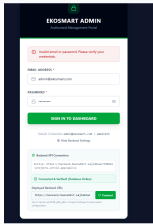


Figure 7.1: Admin Dynamic Form Studio & Field Customizer

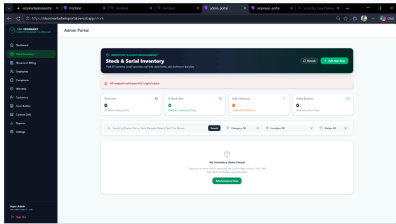


Figure 7.2: Dynamically Rendered Intake Form on Public Website

7.2 Dynamic Field Schema Definition

Property	Type	Description / Operational Role
fieldName	String	Database key used in the complaint's formData JSON payload.
fieldLabel	String	Human-readable label rendered in the user interface.
fieldType	Enum	text, number, select, textarea, date, checkbox, file
required	Boolean	Enforces mandatory validation on client and server before submission.
order	Number	Defines the vertical display order on the customer form.

PAGE 8 — ADMIN EMPLOYEE MANAGEMENT & PERMISSIONS

8.1 Staff Onboarding & Profile Management

The Employee Management module (`admin-portal/src/pages/Employees.tsx`) handles staff profiles, access credentials, department assignments, and security roles. Administrators maintain complete records including Employee ID (EMP-XXXX), name, contact number, official email, department, designation, and password visibility toggles.

[Admin Creates Employee] --> [Assigns Role & Permissions] --> [Saves to MongoDB Employee Model] | [Employee Login] --> [Server Issues JWT with RBAC Payload] <-----+ | v [Client Navigation & API Middleware Enforce Permission Gates]

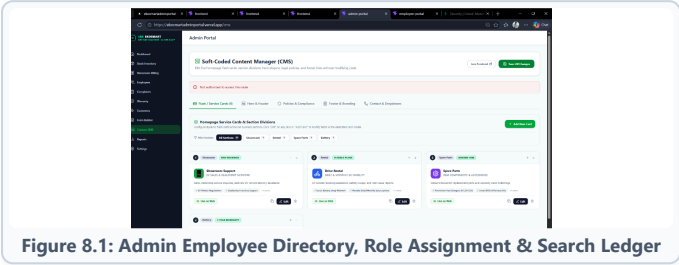


Figure 8.1: Admin Employee Directory, Role Assignment & Search Ledger

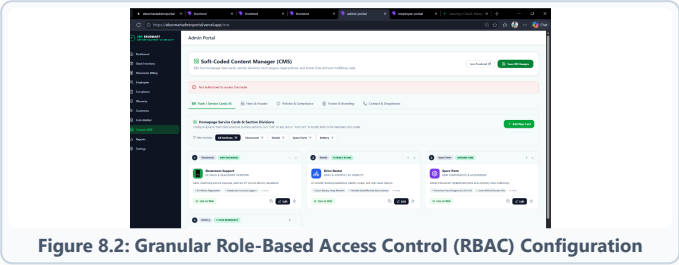


Figure 8.2: Granular Role-Based Access Control (RBAC) Configuration

8.2 Granular RBAC Permission Matrix

Permission Key	Scope & Access Granted	Route Guard / Enforcement
complaints	View & resolve assigned service tickets, log diagnostics	Guards /complaints & /api/v1/complaints/*
customers	Search customer ledger, inspect historical services	Guards /customers & /api/v1/customers/*
warranty	Serial warranty check & claim processing	Guards /warranty & /api/v1/warranty/*
billing	Showroom POS billing counter, tax invoice generation	Guards /billing & /api/v1/billing/*
stock	Warehouse inventory, battery pack lookup, CSV export	Guards /stock & /api/v1/stock/*
scanner	HTML5 camera barcode / QR scanning utilities	Enables scanner modal across modules
reports	View & export assigned workload analytics	Guards /reports & /api/v1/complaints/admin
payslips	View personal monthly remuneration & print payslip	Guards /my-salary & /api/v1/employees/me/payslips

PAGE 9 — ID CARD, SALARY AND PAYROLL

9.1 Digital Employee ID Card Engine

The ID card system (IdCardModal.tsx & MyIdCard.tsx) generates verified digital identification badges featuring company branding, staff photograph/avatar, official Employee ID (EMP-XXXX), division, designation, and an encoded QR badge for on-site scanning.



Figure 9.1: Digital Employee ID Badge with Encoded QR Badge

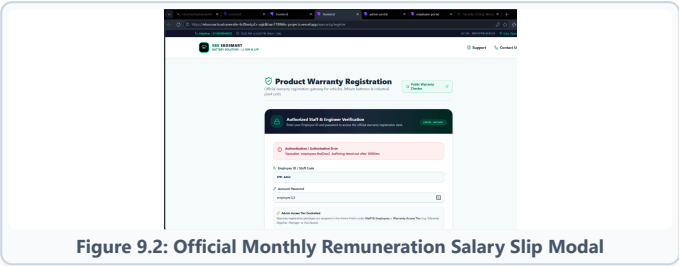


Figure 9.2: Official Monthly Remuneration Salary Slip Modal

9.2 Payroll Calculation & Isolated Payslip Engine

The payroll module (SalarySlipModal.tsx & backend/src/controllers/employee.controller.ts) manages remuneration packages. It records basic earnings, statutory deductions, computes net pay, and translates amounts into Indian Rupees in words.

Earnings & Allowances:

Basic Salary, HRA, Conveyance, Special/Tech Allowance, Overtime & Field Pay, Performance Bonus, Arrears.

Statutory Deductions:

Provident Fund (EPF 12%), ESI Contribution, Professional Tax (PT), TDS / Income Tax, Advance & Loan Recovery.

9.3 Isolated Single-Page A4 Print Engine

The application utilizes an isolated print engine (utils/print.ts). When an employee clicks **Print / Save PDF**, a hidden iframe renders strictly the payslip card, preventing surrounding sidebars, navbars, and buttons from leaking into the printout or PDF.

PAGE 10 — BILLING AND INVOICE SYSTEM

10.1 Showroom Point-of-Sale Billing Terminal

The billing module (BillingManager.tsx & Billing.tsx) serves as the POS checkout system for showrooms and service centers. Operators can add line items, calculate discounts and GST (18%), scan battery pack barcodes, select payment modes (UPI, Cash, Card, Finance), and generate tax invoices.

```
[Select Product / Battery] --> [Scan Serial via Camera Scanner] --> [Validate Stock Level] | [Generate Invoice (INV-YYMM-XXXX)] <-- [Compute GST 18% & Net Total] <-----+ | +----> [Auto-Issues Warranty Record in Database] +----> [Decrements Available Warehouse Stock Quantity]
```

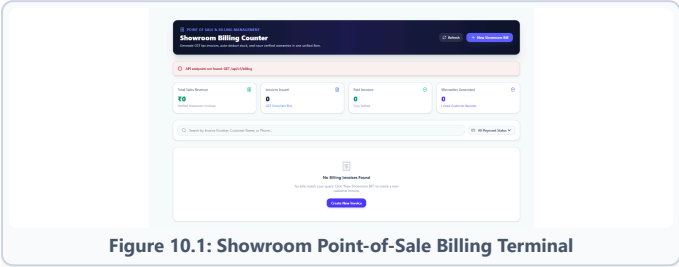


Figure 10.1: Showroom Point-of-Sale Billing Terminal

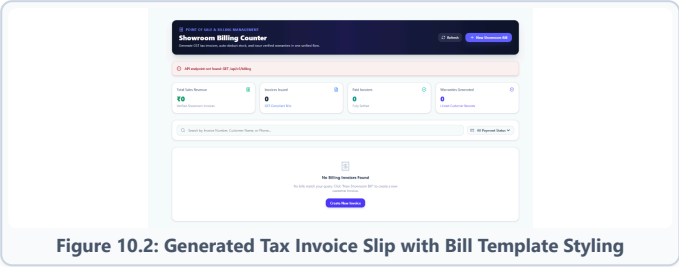


Figure 10.2: Generated Tax Invoice Slip with Bill Template Styling

10.2 Soft-Coded Bill Template Designer

Template Element	Customizable Attributes	Implementation Status
Branding & Theme	Primary highlight color, company logo, watermark text	IMPLEMENTED
Header Profile	Business Name, Tagline, Central Plant Address, GSTIN, Phone	IMPLEMENTED
Footer Disclaimers	Authorized signatory label, terms of return, dispute jurisdiction	IMPLEMENTED
Backend Invoicing API	POST /api/v1/billing & GET /api/v1/billing	IMPLEMENTED

PAGE 11 — EMPLOYEE PORTAL

11.1 Employee Workspace Architecture

The Employee Portal (employee-portal/) provides field engineers and showroom staff with a dedicated workstation. Authentication via `POST /api/v1/auth/login` verifies credentials, issues an `employee_token`, and evaluates RBAC permissions to dynamically configure the workspace.

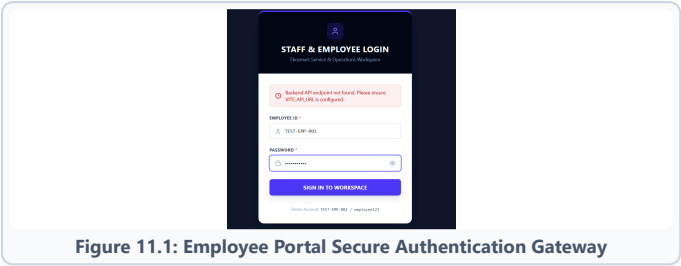


Figure 11.1: Employee Portal Secure Authentication Gateway

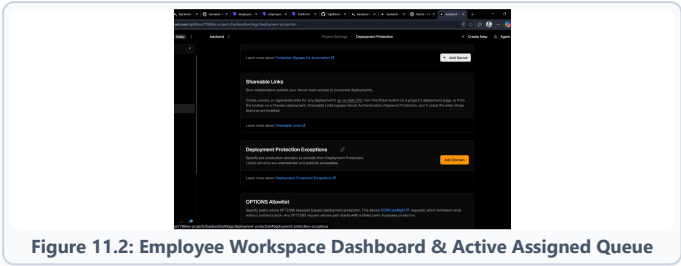


Figure 11.2: Employee Workspace Dashboard & Active Assigned Queue

11.2 Employee Portal Functional Module Directory

Module Page	Route	RBAC Permission	Operational Function
Workspace Dashboard	/	Authenticated Staff	Daily assigned queue summary, task status overview.
Assigned Complaints	/complaints	complaints	Inspect problem dossiers, update status, add repair notes.
Customer Directory	/customers	customers	Search customer profiles and linked service histories.
Warranty Terminal	/warranty	warranty	Check warranty coverage by serial, process service claims.
Showroom Billing	/billing	billing	Generate customer invoices, scan serials, compute GST.
Stock Inventory	/stock	stock	Check warehouse stock levels, scan barcodes, export CSV.
Staff Reports	/reports	reports	Export strictly assigned complaints to CSV.
My Salary Slip	/my-salary	payslips	View monthly remuneration & print official single-page slip.

PAGE 12 — EMPLOYEE COMPLAINT, CUSTOMER & WARRANTY WORKFLOW

12.1 End-to-End Service Resolution Workflow

When an employee logs into the portal, tickets assigned to their division appear in their active queue. Staff open the technical problem dossier (TicketDetailModal.tsx) to inspect customer complaints, photos, and dynamic test metrics.

[Assigned Ticket in Queue] --> [Inspect Customer Details & Photos] --> [Update Status to "IN PROGRESS"] | [Ticket Moved to Permanent History Ledger] <-- [Resolve Ticket & Add Remarks] <--+

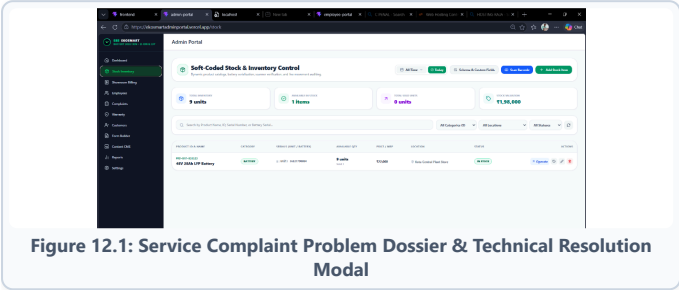


Figure 12.1: Service Complaint Problem Dossier & Technical Resolution Modal

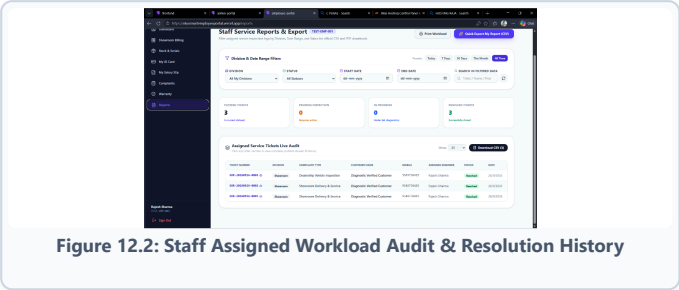


Figure 12.2: Staff Assigned Workload Audit & Resolution History

12.2 Workload Isolation & Active Queue Rules

Operation / Rule	Technical Mechanism	Audit & Ledger Outcome
Assigned Queue Filter	Matches c.assignedTo with user.employeeId and user._id	Staff view only their assigned workload.
Diagnostic Progression	PATCH /api/v1/complaints/:id/status	Updates status from Pending to In Progress to Resolved.
Active vs. History	Resolved/Closed tickets clear from active list	Permanently archived in database for SLA reports.
Warranty Verification	Serial lookup against purchase date	Confirms active coverage before processing free repairs.

PAGE 13 — STOCK, BILLING AND SCANNER

13.1 Warehouse Stock & Inventory Architecture

The inventory engine (backend/src/models/Stock.ts) tracks catalog products, serialized battery packs, and movement ledgers across warehouses and showrooms. Stock items maintain status flags (In Stock, Sold, Reserved, Damaged) and record comprehensive movement histories.

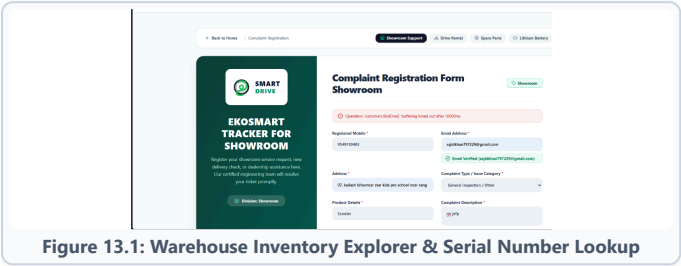


Figure 13.1: Warehouse Inventory Explorer & Serial Number Lookup

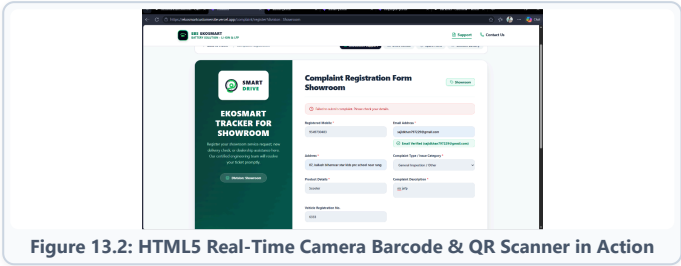


Figure 13.2: HTML5 Real-Time Camera Barcode & QR Scanner in Action

13.2 HTML5 Camera Barcode & QR Scanner Integration

Integrated via ScannerModal.tsx using @zxing/browser, the scanner leverages device webcams or mobile cameras to read Code 128, Code 39, EAN-13, and QR barcodes on physical battery packs, instantly auto-populating line items in POS billing.

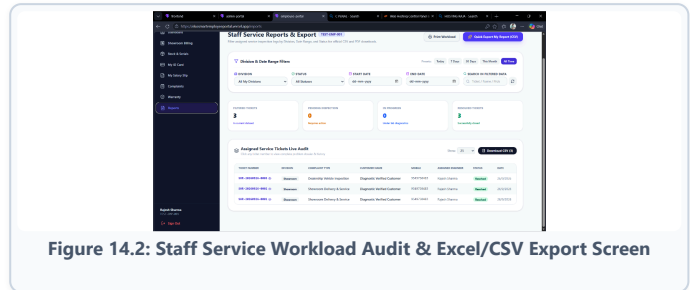
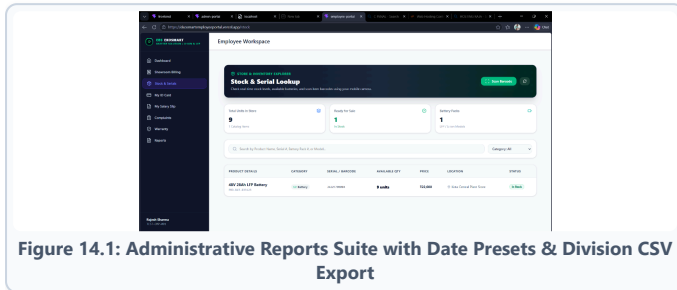
13.3 Billing + Stock Synchronization Workflow

Step	Workflow Phase	System Action / Data Mutation
1	Barcode Scan	Staff scans battery serial; system verifies status === 'In Stock'.
2	POS Invoice Creation	Item added to bill; GST (18%) and grand total computed.
3	Database Sync	Stock.quantity decremented; StockMovement logged as 'Sold'.
4	Warranty Auto-Creation	Warranty record generated linking serial number to customer bill.

PAGE 14 — REPORTS, FILTERS AND EXCEL EXPORT

14.1 Enterprise Reporting & Analytics Suite

The reporting engine (`admin-portal/src/pages/Reports.tsx` & `employee-portal/src/pages/Reports.tsx`) provides analytics across complaints, warranties, inventory, and revenue. It supports division filters (*Battery, Rental, Showroom, Spare Parts*) and date presets.



14.2 Date Range Presets & Workload Isolation

Date Range Filter Presets:

- Today (Default view) & Yesterday
- Last 7 Days & Last 30 Days
- This Month & All Time
- Custom Date Range (Start Date → End Date)

Workload Isolation Guarantee:

Employee Reports strictly filter tickets assigned to the logged-in staff member by matching `employeeId`, `_id`, `name`, and `email`, preventing data leakage across staff.

14.3 Excel / CSV Export Specifications

Export routines format data into RFC 4180 compliant CSV files with \uFEFF UTF-8 Byte Order Marks (BOM), ensuring multilingual customer names and Indian Rupee (&8377;) currency symbols render cleanly in Microsoft Excel and Google Sheets.

PAGE 15 — COMPLETE SYSTEM FLOW & TECHNICAL SUMMARY

15.1 Master End-to-End Operational Flow

[CUSTOMER] --> Public Portal --> Dynamic Form Intake --> Express API --> MongoDB Atlas (Ticket Created) [ADMIN] --> Admin Portal --> CMS / Form Designer / RBAC / Payroll --> MongoDB Atlas (Global Config) [EMPLOYEE] --> Employee Portal --> Diagnostic Dossier / POS / Stock --> MongoDB Atlas (Task Completed)

15.2 Comprehensive System Implementation Audit Matrix

System Module	Verified Status	Technical Architecture & Verification Notes
Public Website	IMPLEMENTED	React 19 SPA, dynamic CMS hero, soft-coded division cards, mobile menu
Admin Portal	IMPLEMENTED	KPI dashboard, CMS studio, dynamic form builder, staff management
Employee Portal	IMPLEMENTED	Role-gated workspace, assigned queue, POS billing, stock lookup
Backend API	IMPLEMENTED	Node.js, Express, TypeScript, dual prefix routing (/api/v1 & /api)
Database Layer	IMPLEMENTED	MongoDB Atlas, Mongoose ODM schemas, serverless reconnect middleware
Authentication & RBAC	IMPLEMENTED	JWT Bearer token, bcrypt password hashing, granular permission flags
Soft-Coded CMS	IMPLEMENTED	Database-driven hero, branding, dynamic service cards, legal policies
Dynamic Form Builder	IMPLEMENTED	Configurable field schema per division with live public form rendering
Complaint System	IMPLEMENTED	Automatic ticket generation (EBS-YYMM-XXXX), lifecycle tracking
Warranty Engine	IMPLEMENTED	Serial registration, automated duration expiry calculation, live search
Customer Directory	IMPLEMENTED	Central customer directory auto-ingested from complaints & invoices
Staff & Permissions	IMPLEMENTED	Staff CRUD, designation/department tracking, granular RBAC flags
Digital ID Cards	IMPLEMENTED	Printable ID card badge with verifiable QR badge and company branding
Salary & Payroll	IMPLEMENTED	Salary structure configuration, allowances, deductions, words INR
Isolated Payslip Print	IMPLEMENTED	Hidden iframe print engine generating single-page A4 payslips / PDF
Showroom POS Billing	IMPLEMENTED	Multi-item invoicing, GST 18%, sequential invoice numbering (INV-XXXX)
Warehouse Stock	IMPLEMENTED	Serialized battery tracking, inventory movement ledger, CSV export
Barcode Scanner	IMPLEMENTED	HTML5 camera scanner (@zxing/browser) with manual entry fallback
Reports & CSV Export	IMPLEMENTED	Multi-tab analytical reports, UTF-8 BOM CSV export, workload isolation
Date Range Filters	IMPLEMENTED	Global preset engine (Today, 7 Days, 30 Days, Month, Custom)
OTP SMS Verification	FRONTEND/UI ONLY	OTP UI dialog present; third-party SMS gateway not connected